SHoEbox - cohort experiment reproduced

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The missing cohort-diversity experiment has been rebuilt and rerun: **12 frozen graphs and 432 fitted corrections**. Validation chooses a nonzero identity correction **3 of 36 times with the small cohort, and 33 of 36 times with the large cohort**. Expanding from 960 to 3,840 training cases improved both private and shared accuracy in 3 of four conditions with the selected identity correction, and 1 of four with GELU.

The repetition matches 80 of 80 published accuracy values at their reported precision. The largest difference from a printed value is 0.004583 percentage points, within its original rounding interval.

Matched protocol

Training coverage	Small	Large
Cases per supported source/query cell	64	256
Total cases	960	3,840
Distinct episode IDs	192	768
Traces excluding seven-byte ID headers	960	3,840
Observed template signatures	869	2,360
Shared / left / right examples	448 / 256 / 256	1,792 / 1,024 / 1,024
Full fact combinations	32	32

All twelve original checkpoints are retained: original/gated publication, dropout/balanced source schedules, and graph seeds 20261106-20261108. The small training cohort is the exact first 64 feature rows of each large-cohort cell. The retained source templates define the template signature, together with source subset, query and the full fact tuple.

Each graph receives two cohort sizes, penalties 0/1/10, identity/GELU activations and initialization seeds 20261121-20261123. Each correction has hidden width 16 and 5,392 parameters. The 16,640-parameter linear base, its normalization, validation (1,920 cases), test/all (384 cases), and independent evaluation (3,456 cases) remain fixed.

Full-batch L-BFGS uses strong Wolfe line search, a 400-iteration cap and L2 coefficient 0.001. Training-only shared-answer margins are capped at 1. Validation chooses strength 0, 0.125, 0.25, 0.5 or 1 by query-balanced NLL, while requiring shared correct counts on all/shared and b_only/shared to meet the frozen base. The same rule chooses between positive penalties 1 and 10; ties prefer smaller strength. Test scores do not select settings.

Selected preservation results

Values are private / shared accuracy in percent, averaged over three graph seeds and three correction initializations. Change is large minus small, in percentage points.

Identity correction

Condition	960 cases	3,840 cases	Change, pp
Original / dropout	73.177 / 87.240	77.257 / 88.108	+4.080 / +0.868
Original / balanced	55.990 / 80.469	57.986 / 82.812	+1.997 / +2.344
Gated / dropout	60.938 / 91.146	59.809 / 90.104	-1.128 / -1.042
Gated / balanced	54.948 / 87.500	56.641 / 88.976	+1.693 / +1.476

GELU correction

Condition	960 cases	3,840 cases	Change, pp
Original / dropout	74.045 / 86.372	75.087 / 83.420	+1.042 / -2.951
Original / balanced	57.248 / 79.080	58.984 / 78.038	+1.736 / -1.042
Gated / dropout	61.024 / 90.625	62.283 / 90.712	+1.259 / +0.087
Gated / balanced	56.033 / 86.372	56.641 / 86.285	+0.608 / -0.087

Nonzero correction selected	960 cases	3,840 cases
Identity	3 / 36	33 / 36
GELU	18 / 36	22 / 36

B-only shared accuracy

Condition	Identity: small / large	GELU: small / large
Original / dropout	86.72 / 90.36	91.32 / 90.10
Original / balanced	100.00 / 100.00	100.00 / 100.00
Gated / dropout	95.83 / 95.57	96.79 / 96.09
Gated / balanced	98.70 / 99.22	98.96 / 98.70

Large-cohort selected GELU missing-information accuracy averages 25.049%, against 25% chance. The default model/readout remains unchanged.

Penalty-zero controls

These use the same fitting, fixed baseline and validation scale selection, with preservation penalty zero.

Identity correction

Condition	960 cases	3,840 cases	Change, pp
Original / dropout	72.917 / 87.500	74.479 / 86.979	+1.562 / -0.521
Original / balanced	55.990 / 80.469	59.071 / 82.031	+3.082 / +1.562
Gated / dropout	60.938 / 91.146	61.719 / 90.104	+0.781 / -1.042
Gated / balanced	54.948 / 87.500	56.380 / 86.198	+1.432 / -1.302

GELU correction

Condition	960 cases	3,840 cases	Change, pp
Original / dropout	73.611 / 87.153	75.000 / 85.590	+1.389 / -1.562
Original / balanced	55.990 / 80.469	59.983 / 78.472	+3.993 / -1.997
Gated / dropout	61.068 / 90.625	62.934 / 90.625	+1.866 / +0.000
Gated / balanced	55.122 / 87.413	55.599 / 87.587	+0.477 / +0.174

Agreement with the surviving report

Published metric group	Values matching precision	Largest difference, pp
selected / identity	16 / 16	0.000500
selected / gelu	16 / 16	0.000472
penalty_0 / identity	16 / 16	0.000500
penalty_0 / gelu	16 / 16	0.000472
B-only shared	16 / 16	0.004583

The reference contains 64 private/shared means rounded to three decimals and 16 B-only means rounded to two. Agreement uses half of the last displayed decimal place. `reported_comparison.json` records every original value, repeated value and difference; `reported_metrics.json` preserves the values transcribed from the historical report.

This rerun uses the preserved protocol and inputs without tuning to the published scores. The unavailable original implementation prevents a byte-level comparison with that original study. The new package supplies a complete, auditable replacement.

Verification and reproducibility

All **307 automated tests pass** across five suites. Nine dedicated cohort tests cover coverage counts, exact nested feature ownership, split separation, frozen normalization, checkpoint restoration and resume rejection after incompatible changes.

The independent artifact audit reconstructs all 432 training objectives, their validation scale/penalty selections, and all 2,880 new prediction groups. It also exactly reproduces 720 prediction groups from the retained residual and emission studies. Saved predictions and discrete decisions compare exactly; named floating loss fields and normalization reconstruction use absolute/relative tolerance 1e-12. Loaded frozen base tensors remain identical.

All input/source hashes, checkpoint lineages, synthetic cases, coverage counts and expert budgets are checked. Training IDs are disjoint from validation and evaluation. The audit checks predictions from saved tensors rather than trusting the reported summary.

432 of 432 fits reached the 400-iteration cap. Final maximum gradient magnitudes range from 2.394007e-06 to 0.02225517. The largest feature difference from the legacy small training cache is 0.0003931746; both newly fitted cohort sizes use the same fresh capture, while retaining the original base normalization.

Training capture used **96,768,000 expert executions**, with K=1 and 84 per event. Fitting used cached features. Runtime: PyTorch 2.15.0+cpu, two threads per worker and four fitting workers. Every fitted correction is checkpointed atomically. Resume verifies the exact source, configuration, inputs, capture and base before continuing.

Downloadable package

`SHoEbox_cohort_reproduction_source.zip` contains the source, reports, aggregate audit and verification logs. `SHoEbox_cohort_reproduction_artifacts.zip` additionally contains every required graph, inherited readout, feature/validation cache, all 432 newly fitted corrections, selections and predictions. Each archive has a file-by-file SHA-256 manifest, its source commit and publication status.

From the extracted artifact package:

```
python -m pip install -r requirements.txt
python scripts/check_all.py
python -m experiments.text_regions.diversity_compare \
  --source artifacts/cohort_reproduction_v1 \
  --out artifacts/reproduced_cohort_comparison.json
```

For a fresh fit, run `scripts/run_diversity_study.py` with a new output directory, `--jobs 4 --threads 2`. To continue an interrupted fit, add `--resume` with its existing output directory.

The next development step remains a full-rank linear decoder refit on the expanded cohort, compared on the same fixed evaluations and measured for inference cost. That follow-up is separate from this completed reproduction.